

B.Tech. Degree IV Semester Examination April 2014**CS 1404 AUTOMATA LANGUAGES AND COMPUTATIONS**
(2012 Scheme)

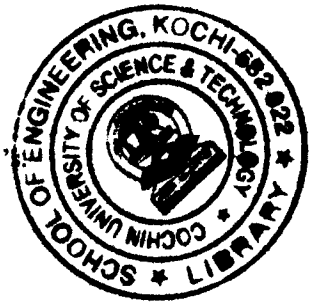
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Differentiate between DFA and NFA.
- (b) Write a note on NFA with null moves.
- (c) Write a note on ambiguous grammar. Give an example.
- (d) Define acceptance of a string by push down automata.
- (e) Design a turing machine to recognise all strings consisting of even number of 1's.
- (f) What is non deterministic turing machines?
- (g) Write a note on linear bounded automata.
- (h) Write a short note on Chomsky classification of languages.

**PART B**

(4 x 15 = 60)

- II. Prove the equivalence of DFA and NFA. Give an example. (15)
- OR**
- III. (a) Write the procedure of minimization of automata with an example. (10)
- (b) Explain pumping lemma for regular grammar and its applications. (5)
- IV. (a) Write notes on simplification of CFG (8)
 - (i) removing null productions
 - (ii) removing unit productions
- (b) Explain with example (7)
 - (i) Chomsky normal form
 - (ii) Greibach normal form
- OR**
- V. (a) Explain PDA and DPDA, with a neat diagram. (7)
- (b) Construct a PDA accepting the set of all strings over {a,b} with equal number of a's and b's. (8)
- VI. Explain any two techniques for constructing a turing machine. (15)
- OR**
- VII. (a) Write a short note on universal turing machines. (8)
- (b) Design a turing machine to recognize the language $\{1^n 2^n 3^n \mid n \geq 1\}$. (7)
- VIII. (a) Explain the halting problem of turing machine. (7)
- (b) Write a short note on recursively enumerable languages. (8)
- OR**
- IX. Explain conversion of regular grammar to finite automata, with the help of suitable examples. (15)